A4 HAND-DRAWING REFERENCE

Whale Watching Around Australia

Seasons, species and the habitats worth protecting

Draw this layout yourself. All chart shapes below are schematic, not results.

1 FIND THE WHALES

Introduce the observations and their uneven coverage.

01 A

Where are whales recorded?

Point symbol map



Position = record; colour = species.

02 A

Where do records cluster?

Proportional symbol map

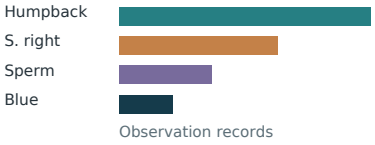


Circle area = records per 1-degree cell.

03 A

Which species are reported?

Ranked bar chart



Length = record count; show sample sizes.

2 FOLLOW THE SEASONS

Compare recorded timing with documented habitat seasons.

04 A

When do regions report whales?

Month-by-coast heatmap

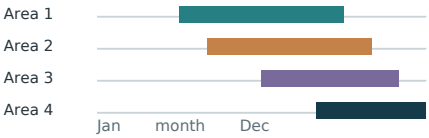


Colour = within-coast monthly share.

05 B

When is habitat used?

Season interval chart



Span = documented months at named areas.

06 A

Do reported latitudes shift?

Monthly latitude box plot



Humpbacks only; box = middle half.

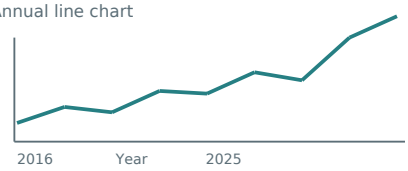
3 READ THE EVIDENCE

Explain species differences and limits of citizen observations.

07 A

How has reporting changed?

Annual line chart

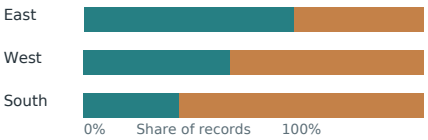


2016-2025 only; this is reporting effort.

08 A

How does species mix vary?

100% stacked bar chart



Same species colours; display each total.

09 A

Which years dominate the data?

Year/species treemap



Area = records; nesting = year then species.

4 WATCH WITH CONSERVATION IN MIND

Connect migration and habitat use to responsible viewing.

10 B

Why protect whale corridors?

Schematic migration flow map

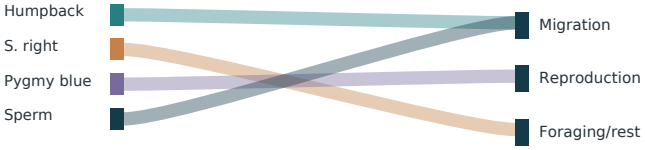


Arrows from BIA descriptions; no inferred tracks.

11 B

What do whales use habitats for?

Species-to-behaviour alluvial



Width = unique BIA records; not whale movements.

A = iNaturalist observations via OBIS B = DCCEEW BIAs / Footer: sources, author, date, AI credit

VERIFIED DATA SOURCES

Two datasets you can use

The supplied files are compact extracts of real records. Use the observations for comparison and the government habitat data for ecological context.

A iNaturalist marine observations

3,082 records for four whale species in a geographic window around mainland Australia and Tasmania. Extract dates: 2016-06-08 to 2026-07-19. Source release: 1 September 2026. Main file: 01_whale_observations.csv.

Blue whale	**7 records**
Humpback whale	**2,726 records**
Southern right whale	**300 records**
Sperm whale	**49 records**

Use species, event date, latitude/longitude and supplied state for maps, monthly comparisons and reporting patterns. Humpbacks and southern right whales should lead the story. Seven blue-whale observations are too sparse for a reliable seasonal pattern.

Interpretation: these are community occurrence records, not a whale census, tourist counts or guaranteed live encounters. The file flags possible non-live mentions for review. Keep 2026 separate because it is incomplete. Preserve Unassigned locations instead of guessing their state.

Open source dataset | DOI and citation | Collection licence: CC BY-NC 4.0. Preserve record-level credits.

B Whale Biologically Important Areas

44 unique BIA records from DCCEEW whale layers, queried 15 October 2026. Files: 02_whale_bia.geojson for maps and 02_whale_bia_attributes.csv for tables. These are two formats of the same source.

Fields include species, named location, region, dominant behaviour, seasonal descriptions and source/review information. Categories cover humpback, southern right, pygmy blue and sperm whales. Use the selected season-window CSV to start a carefully labelled seasonal interval chart.

Interpretation: BIAs identify important habitat; they are not automatically legally protected areas or complete species ranges. Geometry in this pack is simplified for overview maps. Underlying evidence dates vary. A blank season is unspecified, not absence.

Official BIA explanation | Official download portal | Licence: CC BY; credit Commonwealth of Australia / DCCEEW.

Scope decision: focus on seasons, species and conservation. Neither source measures whale-watching visitors or revenue. Do not turn observation growth into a tourism-growth claim.

Locations and seasonal comparisons

Each panel has a specific question, a source and a measurable encoding. Use these notes when drawing and when writing the template's How section.

01 Point symbol map

A: longitude, latitude, species. One small, semi-transparent dot per record. Filter by species and year; tooltip shows date and source link. Name this recorded observations, not guaranteed viewing locations.

02 Proportional symbol map

A: aggregate records into fixed 1-degree cells. Circle AREA, not radius, represents count. Let the reader compare concentrations for a selected season. Counts reflect reporting intensity; this is not population density.

03 Ranked species bars

A: count distinct occurrence IDs by species. Begin bars at zero and print counts. Keep the two sparse species visible but clearly labelled. Do not interpret counts as relative abundance.

04 Monthly reporting heatmap

A: month by supplied-state coastal group. Colour = percentage of each coast group's records falling in each month, with n in tooltips. Show missing state separately. Use full years for comparisons; unsurveyed is not absent.

05 Habitat season intervals

B: named BIA, species and documented months. Start with 02_selected_season_windows.csv. Show approximate windows and preserve original notes in tooltips. Split a season crossing December/January. Unknown stays unknown.

06 Humpback latitude box plots

A: monthly latitude distributions for humpback records only. Explain the box as the middle half of recorded locations and the line as the middle location. Suppress groups with fewer than 20 records. This does not track individual migration.

Map design: use a consistent Australian Albers equal-area projection centred around 132 E, 28 S, with parallels near 18 S and 36 S. Provide a coastline basemap and credit its source. It is a supporting geographic layer, not another whale dataset.

Evidence and conservation

Keep narrative text short. After computing the charts, replace neutral questions with accurate findings supported by the displayed data.

07 Annual reporting line

A: number of records by full calendar year, 2016-2025. The title must say observation reporting. Annotate changing participation and data coverage; no population recovery or tourist-growth claim. Show 2026 separately, if used.

08 Species composition bars

A: 100% stacked bars for East, West and South groups, plus Unassigned where useful. Use stable species colours and print total n. This compares the mix of submitted records, not the actual mix of whales in the sea.

09 Year and species treemap

A: nest species inside reporting year; rectangle area = record count. This exposes which years and species dominate the evidence. Use Vega for the hierarchy. Label small groups directly or in tooltips; do not add many tiny categories.

10 Migration and habitat map

B: show habitat polygons for a chosen species with clearly schematic direction arrows based on MIGRATION/SEASON text. Use equal-width arrows: no animal volumes are available. Never connect independent sightings into routes. Add short callouts for resting, reproduction and feeding habitat.

11 Species and habitat-use alluvial

B: link COMMON_NAME to BEHAVIOUR_CAT. Width = number of unique OBJECTIDs in each link. Use Vega; explain that these are mapped BIA records, not animals, habitat area or an importance ranking. Retain pygmy blue as a distinct BIA label.

How to turn page 1 into your assessed sketch

On a blank A4 sheet, draw the title and four section bands. Copy the chart boxes, small chart shapes, axis/legend labels and interaction notes. Add one short narrative sentence per section and a footer for sources, author, date and AI acknowledgement. Photograph or scan your own drawing and save it as a PDF.

The plan has 11 panels, including 3 map idioms and 8 idioms outside the rubric's basic list. This is a design target, not a promised grade. Ask your tutor to confirm the chart count and whether overlapping comparisons are sufficiently distinct. If panel 09 feels repetitive after analysis, redesign it before finalising the webpage.

Your submitted webpage must follow your hand-drawn sketch, use Vega/Vega-Lite, fit one scrolling page and include readable JSON in your public GitHub repository. The rubric gives zero for a digitally created sketch. This PDF is a drawing reference, not a substitute for that submission.